International Economic Development

ECON 4660/8666

Lecture 4. Inequality and poverty

Main Questions:

- How can we best measure inequality and poverty?
- What is so bad about extreme inequality? What is the extent of relative inequality in developing countries; how is this related to the extent of poverty?
- Who are the poor, and what are their economic characteristics?
- What kinds of policies are required to reduce the magnitude and extent of absolute poverty?

I. Growth and the living standards:

1. Growth is a necessary condition for improving living standards.
   - Economic growth can make some or all people better off without making anyone worse off. (Pareto improvement)

2. Growth is not a sufficient condition for improving living standards.

2. Growth is not a sufficient condition for improving living standards.

A. The increased output is used to increase the power of the rulers.

B. Increased output is used for reinvestment.

C. The rich gets richer, the poor gets poorer.

https://www.youtube.com/watch?v=hVimVzgtD6w

II. Measuring Inequality and Poverty

1. Income distribution.

A. Functional distribution of income. Income distributed according to the function performed by the income receivers.

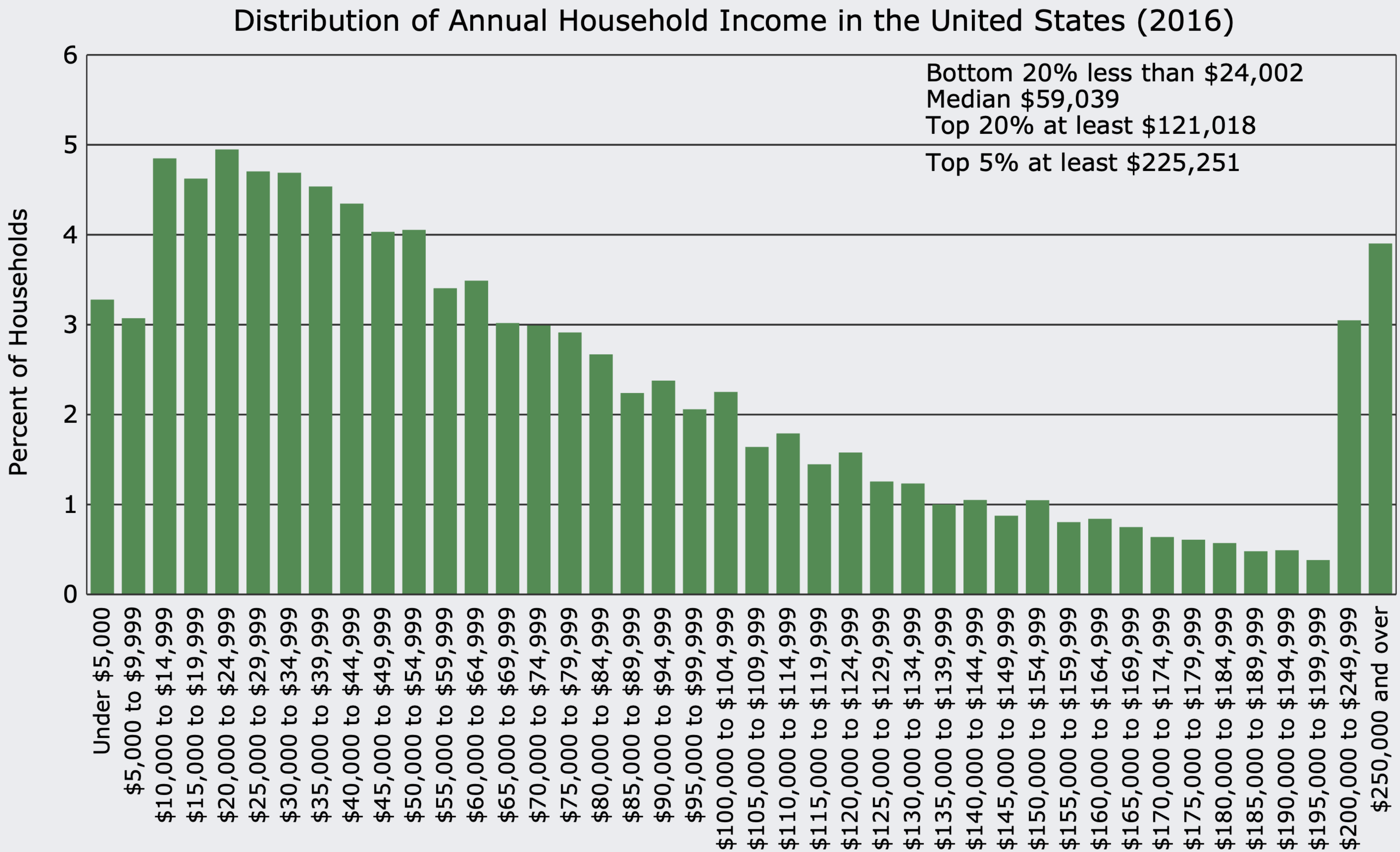
Labor	75%
Capital	15%
Others	10%

B. Personal distribution of income.

Income distributed among households.

Poorest 20% …… 5%

Richest 20% …… 45%



Source: U.S. Census Bureau, Current Population Survey, 2017 Annual Social and Economic Supplement

Frederick E. Allen, Forbes
http://www.forbes.com/sites/frederickallen/2012/10/02/how-income-inequality-is-damaging-the-u-s/

Income Distribution

	percentage of people	percentage of income	cumulative % of people from poor to rich	cumulative % of income
Poorest	**25%**	**10%**	**25%**	**10%**
	25%	**15%**	**50%**	**25%**
	25%	**25%**	**75%**	**50%**
Richest	**25%**	**50%**	**100%**	**100%**

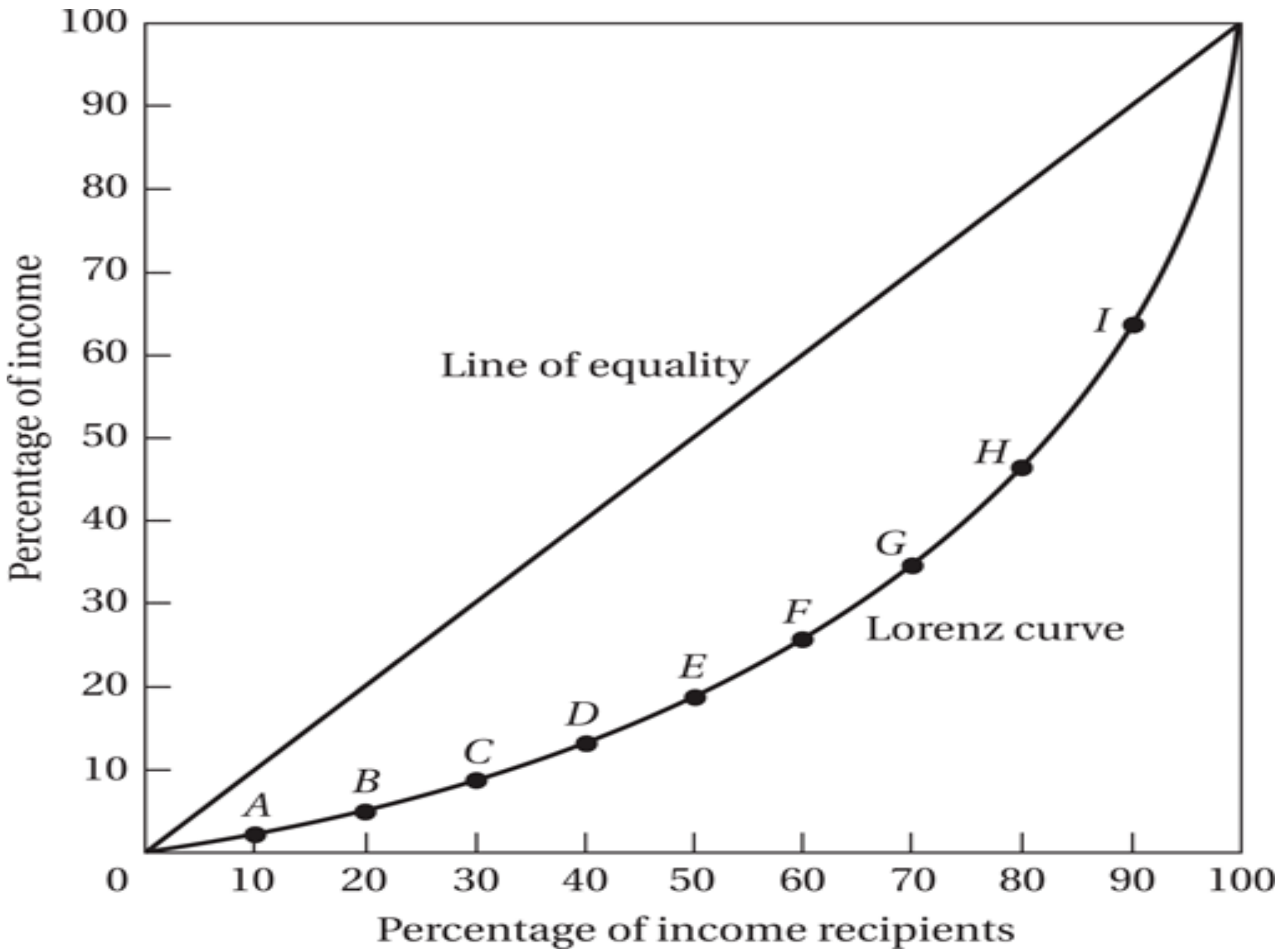
Typical Size Distribution of Personal Income in a Developing Country by Income Shares—Quintiles and Deciles

Individuals	Personal Income (money units)	Share of Total Income (%) Quintiles	Deciles
1	0.8		
2	1.0		1.8
3	1.4		
4	1.8	5	3.2
5	1.9		
6	2.0		3.9
7	2.4		
8	2.7	9	5.1
9	2.8		
10	3.0		5.8
11	3.4		
12	3.8	13	7.2
13	4.2		
14	4.8		9.0
15	5.9		
16	7.1	22	13.0
17	10.5		
18	12.0		22.5
19	13.5		
20	15.0	51	28.5
Total (national income)	100.0	100	100.0

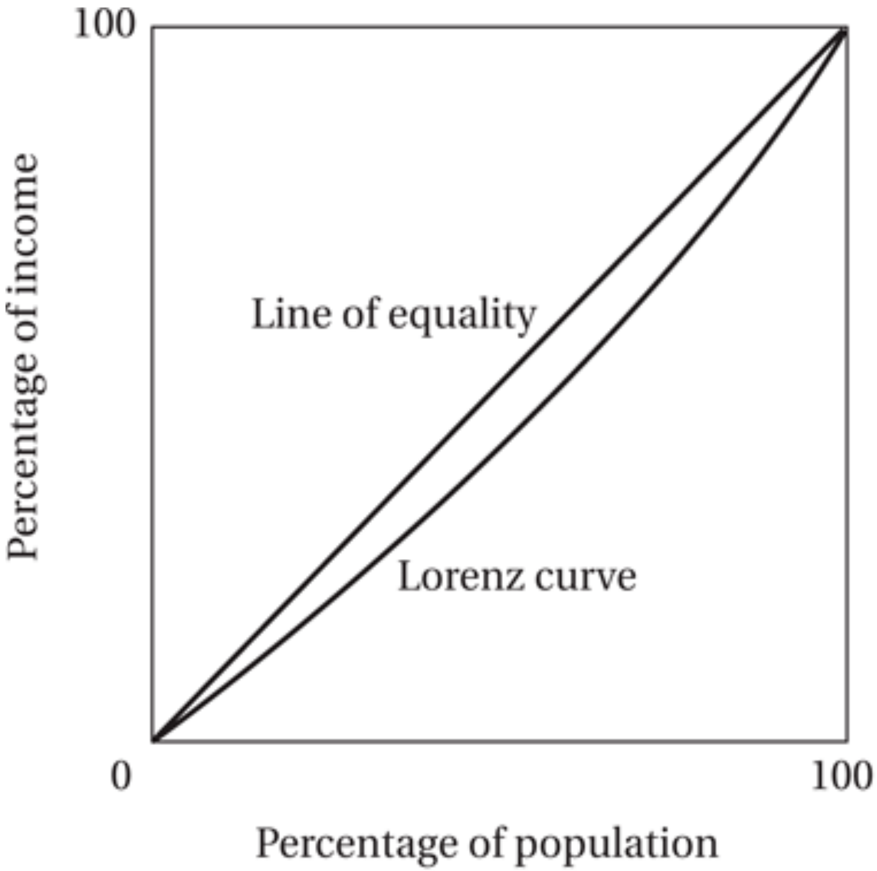
Table 4.5. Alternative measures of progressivity of taxes in selected OECD countries, mid-2000s			
	B. Percentage share of richest decile		
	1. Share of taxes of richest decile	2. Share of market income of richest decile	3. Ratio of shares for richest decile (1/2)
Australia	36.8	28.6	1.29
Austria	28.5	26.1	1.10
Belgium	25.4	27.1	0.94
Canada	35.8	29.3	1.22
Czech Republic	34.3	29.4	1.17
Denmark	26.2	25.7	1.02
Finland	32.3	26.9	1.20
France	28.0	25.5	1.10
Germany	31.2	29.2	1.07
Iceland	21.6	24.0	0.90
Ireland	39.1	30.9	1.26
Italy	42.2	35.8	1.18
Japan	28.5	28.1	1.01
Korea	27.4	23.4	1.17
Luxembourg	30.3	26.4	1.15
Netherlands	35.2	27.5	1.28
New Zealand	35.9	30.3	1.19
Norway	27.4	28.9	0.95
Poland	28.3	33.9	0.84
Slovak Republic	32.0	28.0	1.14
Sweden	26.7	26.6	1.00
Switzerland	20.9	23.5	0.89
United Kingdom	38.6	32.3	1.20
United States	45.1	33.5	1.35
OECD-24	31.6	28.4	1.11
Source: Computations based on OECD income distribution questionnaire.			

http://gregmankiw.blogspot.com/2011/03/what-nation-has-most-progressive-tax.html

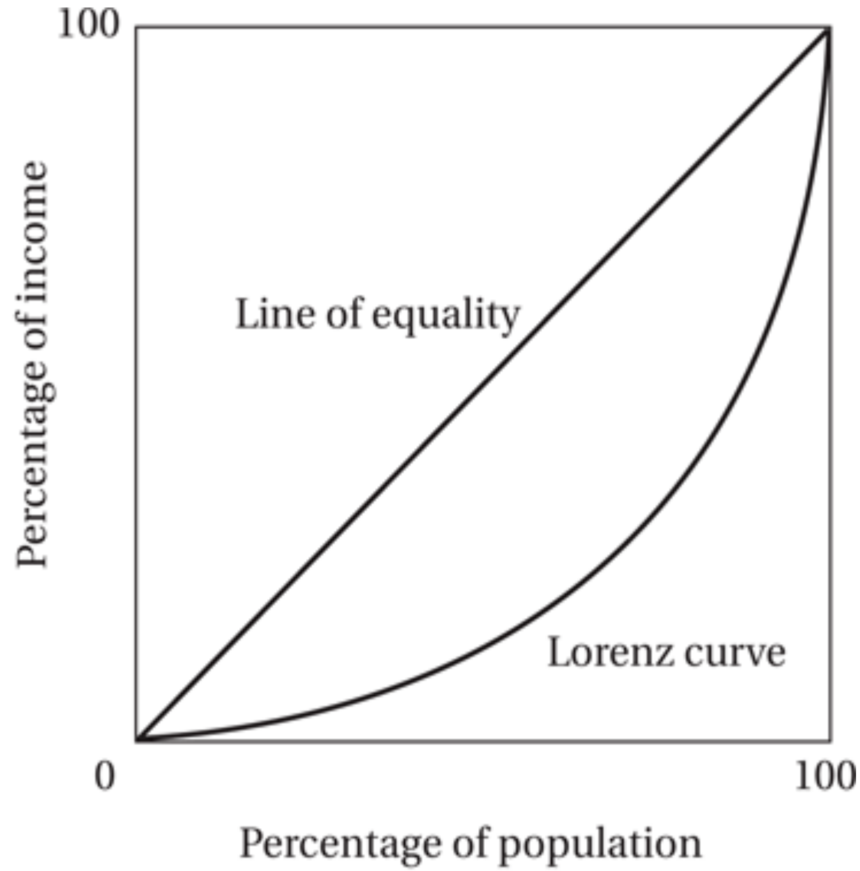
The Lorenz Curve



The Greater the Curvature of the Lorenz Line, the Greater the Relative Degree of Inequality



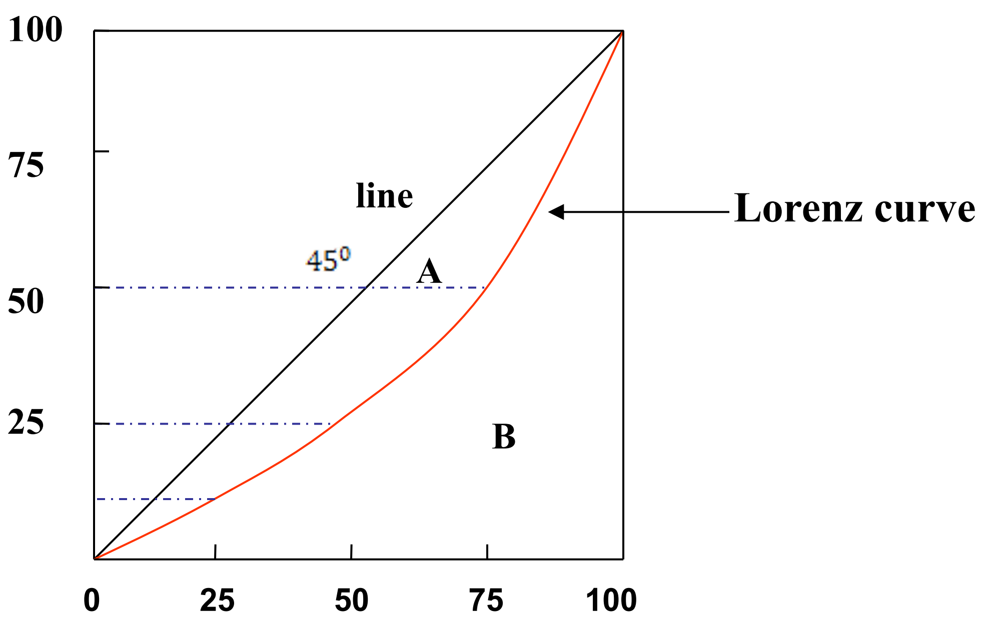
(a) A relatively equal distribution



(b) A relatively unequal distribution

2. Inequality

- Gini concentration ratio: A single number indicating overall inequality.

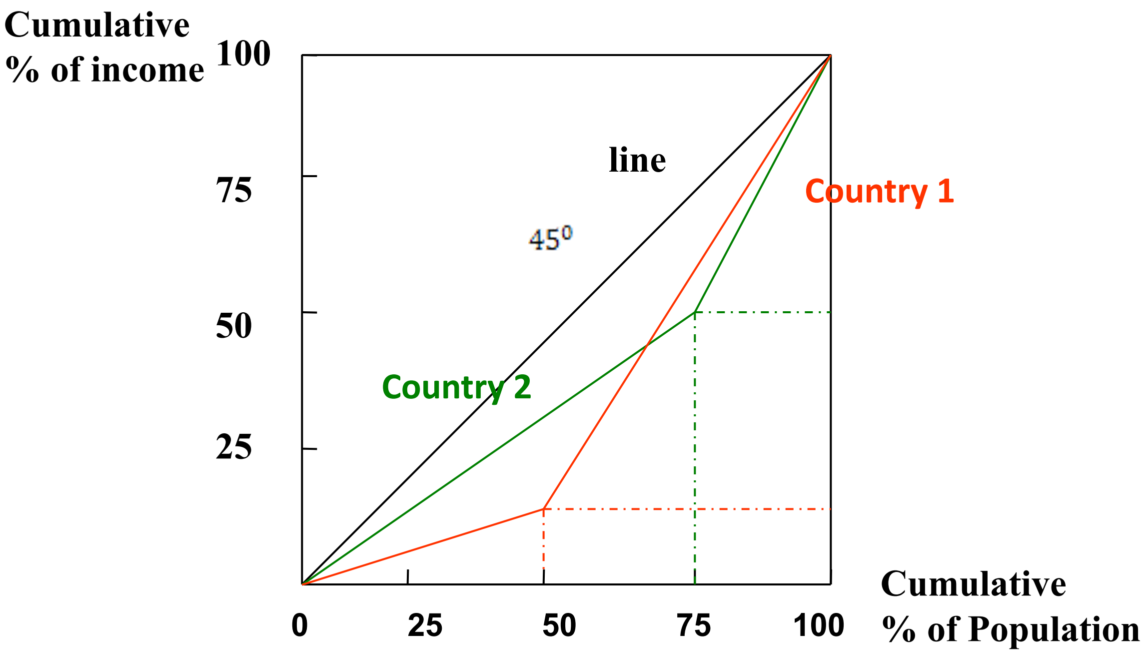


$$\frac{A}{A+B} = Gini\ concentration\ ratio\ (or\ coefficient)$$

Same Gini coefficient may represent different income distribution.

Example:

		Country 1	Country 2
Lowest	**25%**	**5**	**15**
Next	**25%**	**5**	**15**
Next	**25%**	**45**	**15**
Highest	**25%**	**45**	**50**



- Gini coefficient:

Country 1:

Area A: 5000- [(50×10) ÷2 +50×10+(50×90) ÷2] = 5000-3000=2000

$$Gini\ Coef. = \frac{2000}{5000} = 0.4$$

Country 2:

Area A: 5000- [(75×45) ÷2 +25×45+(25×55) ÷2] = 5000-3500=1500

$$Gini\ Coef. = \frac{1500}{5000} = 0.3$$

Income GINI Coefficients in Selected Regions (2000-2010)

Country	GINI	Country	GINI
Hungary	30.0	USA	40.8
Japan	24.9	China	41.5
Sweden	25.0	Singapore	42.5
Germany	28.3	Russia	43.7
France	32.7	Hong Kong	43.4
UK	36.0	Mexico	51.6
India	36.8	Brazil	55.0

Sources：United Nations: *Human Development Report* 2010

3. Poverty

- Desirable properties for poverty measures:
  - Anonymity
  - Population independence
  - Monotonicity
  - Distributional sensitivity

- A condition below the minimum acceptable standard.
- Poverty line measured by income level.
  - Example: The official poverty level for a family of three $ 13,860 in 2000 in the U.S..

    $ 15,720 in 2005

    $ 18,310 in 2009 22,890 Alaska 21,060 Hawaii

    $ 19,090 in 2012 23,870 21,960

    48 States and District Columbia
  - The official poverty level are updated each year to reflect the price change.

- The US Census declared that, in 2011, the official poverty rate was 15.0 percent. There were 46.2 million people in poverty.

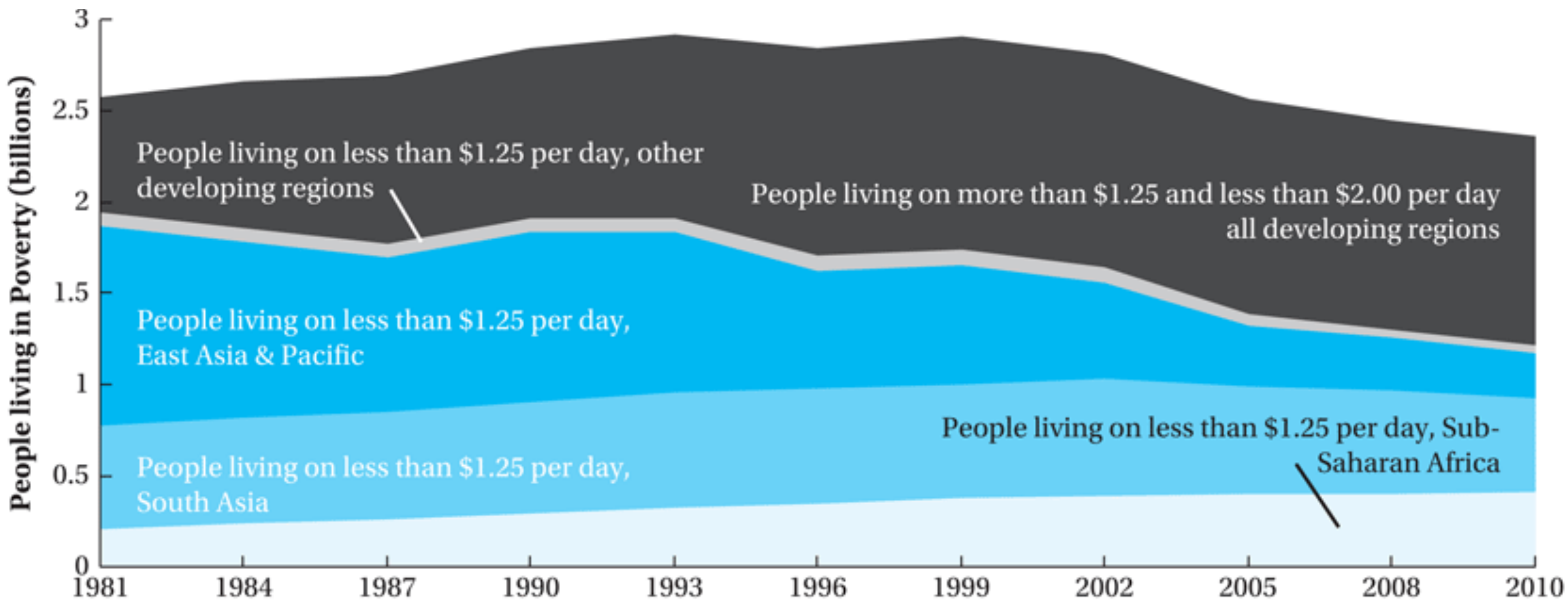
- The poverty rate in 2011

  for children under age 18 was 21.9%

  for people aged 18 to 64 was 13.7%

  for people aged 65 and older was 8.7%

Global and Regional Poverty Trends, 1981–2010



Source: Figure drawn using data from PovcalNet/World Bank; data downloaded 13 February 2014 from http://iresearch.worldbank.org/PovcalNet/index.htm?1.

Measuring Absolute Poverty

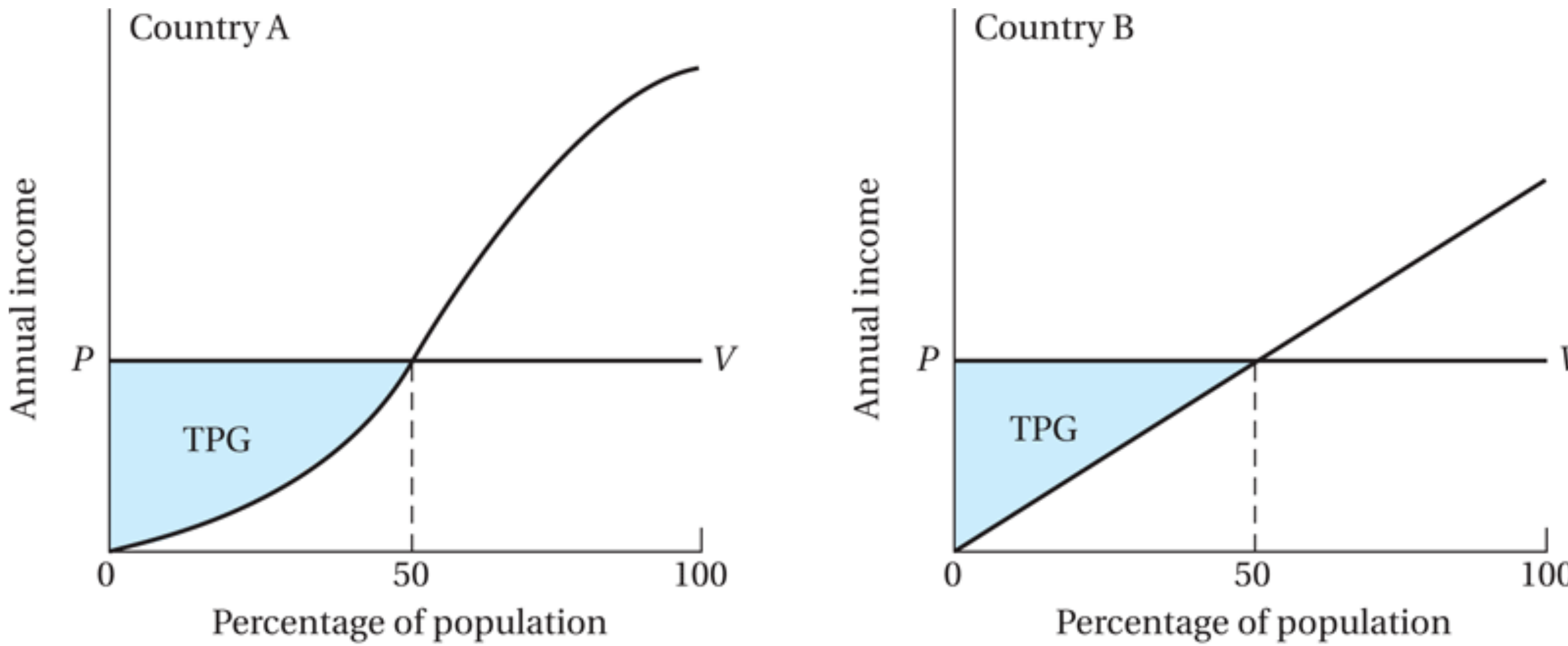
- Headcount Index: *H/N*
  - where H is the number of persons who are poor, and N is the total number of people in the economy

- Total poverty gap:

$$TPG = \sum_{i=1}^{H} (Y_p - Y_i)$$

  - where Y_p is the absolute poverty line; and Y_i the income of the ith poor person

Measuring the Total Poverty Gap



(a) A relatively large poverty gap

(b) A relatively small poverty gap

Measuring Absolute Poverty

- Average poverty gap (APG):

$$APG = \frac{TPG}{N}$$

  - Where N is number of persons in the economy
  - TPG is total poverty gap
  - Note: normalized poverty gap, $NPG = APG/Y_p$

Measuring Absolute Poverty

- Measuring Absolute Poverty
  - Average income shortfall (AIS):

$$AIS = \frac{TPG}{H}$$

  - Where H is number of poor persons
  - TPG is total poverty gap
  - Note: Normalized income shortfall, $NIS = AIS / Y_p$

Measuring Absolute Poverty

- The Foster-Greer-Thorbecke (FGT) index:

$$P_\alpha = \frac{1}{N} \sum_{i=1}^{H} \left(\frac{Y_p - Y_i}{Y_p} \right)^{\alpha}$$

  - N is the number of persons, H is the number of poor persons, and $\alpha \geq 0$ is a parameter
  - When $\alpha=0$, we get the headcount index measure
  - When $\alpha=2$, we get the "P_2" measure

Regional Poverty Incidence, 2010

Region	Headcount Ratio (P_0)	Poverty Gap (P_1)	Squared Poverty Gap (P_2)
Regional Aggregation at $1.25 per Day			
East Asia and the Pacific	12.48	2.82	0.93
Europe and Central Asia	0.66	0.21	0.13
Latin America and the Caribbean	5.53	2.89	2.12
Middle East and North Africa	2.41	0.55	0.23
South Asia	31.03	7.09	2.36
Sub-Saharan Africa	48.47	20.95	11.85
Total	20.63	6.3	2.92
Regional Aggregation at $2 per Day			
East Asia and the Pacific	29.14	9.42	4.05
Europe and Central Asia	2.27	0.64	0.3
Latin America and the Caribbean	10.18	4.67	3.13
Middle East and North Africa	11.55	2.66	0.99
South Asia	65.8	22.86	10.19
Sub-Saharan Africa	69.31	35.22	22.03
Total	40.08	15.32	7.79

Source: data from World Bank, "PovcalNet," http://iresearch.worldbank.org/PovcalNet, accessed 13 February 2014.

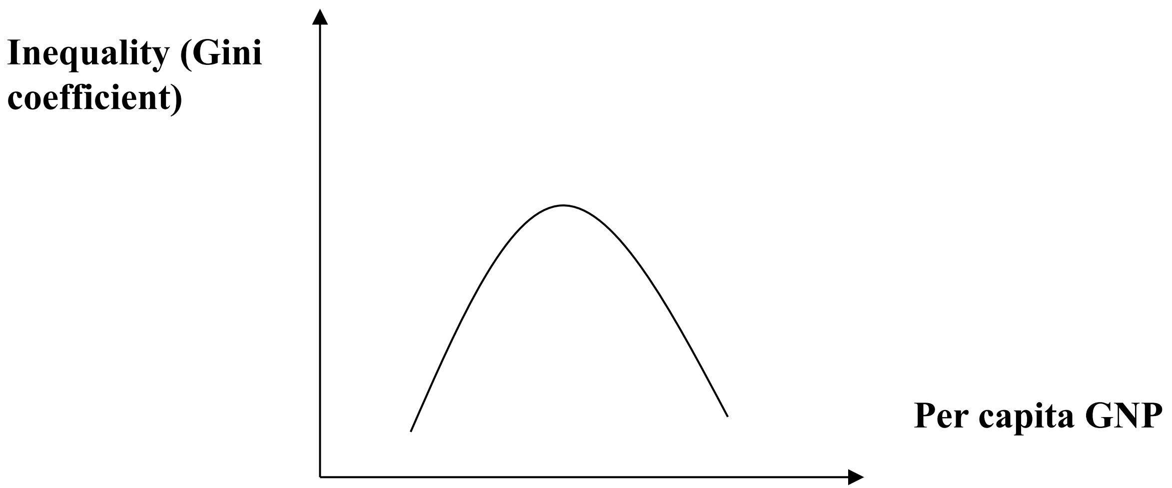
III. Poverty, inequality and social welfare

Why should inequality be a concern?

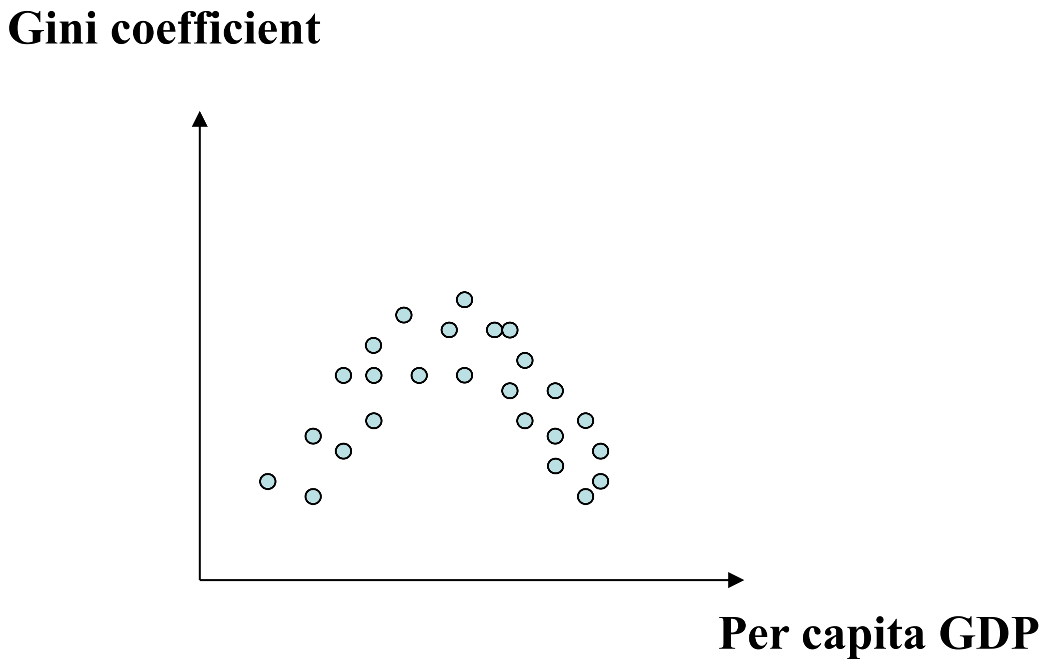
- Extreme income inequality leads to economic inefficiency.
  - Financial constraint, capital flight, limited saving / investment.
- Extreme income disparities undermine social stability and solidarity.
  - Rent seeking
- Extreme inequality if generally viewed as unfair.

1. Kuznets' Inverted U.

– Simon Kuznets, 1955 " Economic Growth & Income Inequality", AER 45, PP.1-28. Date from U.S., U.K. & Germany.



– As GNP ↑, Inequality initially ↑, then reach maximum, then↓.
– This theory received some support, especially from cross - section analyses. (or cross - country analyses)

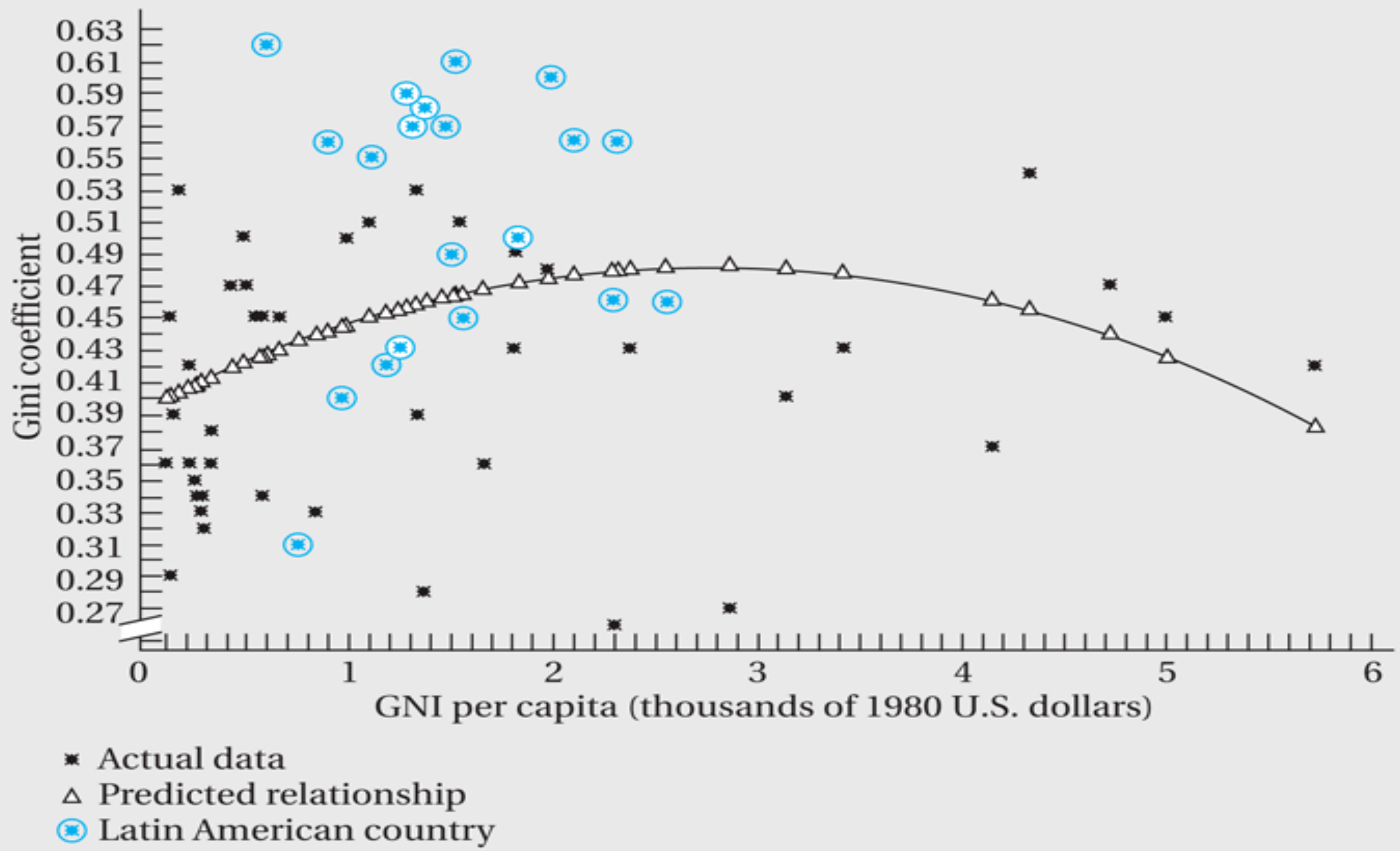


$$G = \alpha_0 + \alpha_1 \ln Y + \alpha_2 (\ln Y)^2$$

$$\alpha_1 > 0, \alpha_2 < 0$$

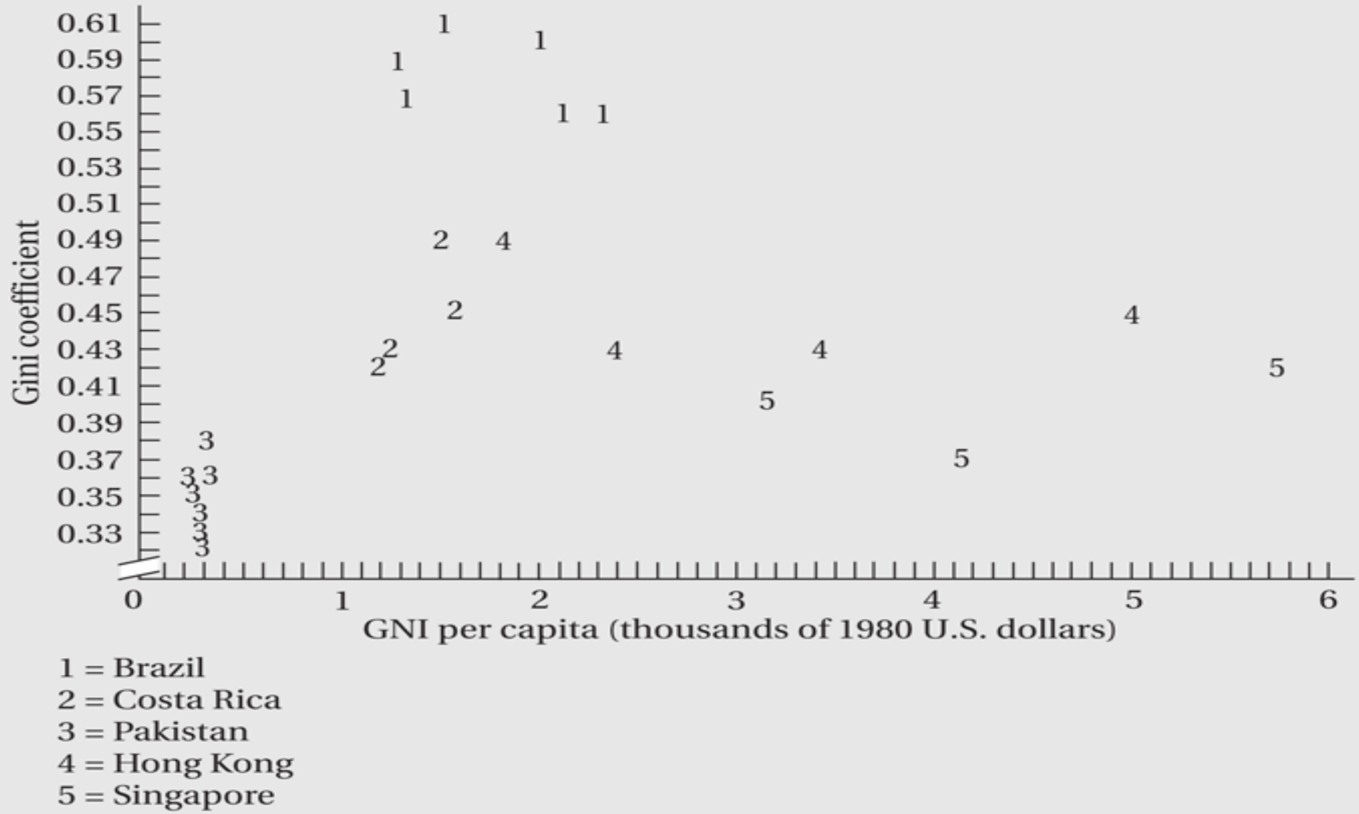
- Klaus Deininger & Lyn Squir, 1996
  - "Measuring Income Inequality: A New Data Base," *World Bank Econ. Review* 10, 1996.
  - Little evidence of a Kuznets Curve in the data. 88 countries with decade- long growth, inequality improved in 45 and worsened in 43. In most cases, the changes in inequality were relatively small & uncorrelated with initial income.

Kuznets Curve with Latin American Countries Identified



Source: Gary S. Fields, *Distribution and Development: A New Look at the Developing World* (Cambridge, Mass.: MIT Press, 2001), ch. 3, p. 46. © 2001 Massachusetts Institute of Technology, by permission of The MIT Press.

Plot of Inequality Data for Selected Countries



Source: Gary S. Fields, *Distribution and Development: A New Look at the Developing World* (Cambridge, Mass.: MIT Press, 2001), ch. 3, p. 44.

Income and Inequality in Selected Countries

Country	Income Per Capita (U.S. $, 2008)	Gini Coefficient	Survey Year for Gini Calculation
Low Income			
Ethiopia	280	29.8	2005
Mozambique	380	47.1	2003
Nepal	400	47.3	2004
Cambodia	640	40.7	2007
Zambia	950	50.7	2005
Lower Middle Income			
India	1,040	36.8	2005
Cameroon	1,150	44.6	2001
Bolivia	1,460	57.2	2007
Egypt	1,800	32.1	2005
Indonesia	1,880	37.6	2007
Upper Middle Income			
Namibia	4,210	74.3	1993
Bulgaria	5,490	29.2	2003
South Africa	5,820	57.8	2000
Argentina	7,190	48.8	2006
Brazil	7,300	55.0	2007
Mexico	9,990	51.6	2008
Upper Income			
Hungary	12,810	30.0	2004
Spain	31,930	34.7	2000
Germany	42,710	28.3	2000
United States	47,930	40.8	2000
Norway	87,340	25.8	2000

Source: data from World Bank, *World Development Indicators, 2010* (Washington, D.C.: World Bank, 2010), tabs. 1.1 and 2.9.

2. The Effect of Inequality on Income & Growth

A. Inequality, Saving, and Capital Accumulation.

- The rich have lower marginal propensity to consume,

  Greater inequality leads to greater savings, and thus greater capital accumulation & growth.
- Friedman (1957) showed that consumption depends on permanent income, & MPC is constant. Thus, distribution of income has no effect on saving.
- Evidence from East Asia shows no relationship between inequality & savings.

B. Inequality, Fiscal policy, and Growth.

- Greater inequality leads to higher taxes & low growth.
- Alesina & Rodrik (1994), “Distributive Politics & Economic Growth”, QJE 109, 465-90.

C. Inequality & Political Stability

- Greater inequality leads to more political instability and lower growth.
- Overall, there is no evidence about the relationship between inequality & growth from time–series data for individual countries.
- There exists cross-country evidence that inequality and growth are negatively related.

IV. Characteristics of high-poverty groups

- Rural poverty
- Women and poverty
- Ethnic minorities, indigenous populations and poverty

Poverty: Rural versus Urban

Region and Country	Survey Year	Percentage below National Poverty Line: Rural Population	Urban Population	National Population
Sub-Saharan Africa				
Benin	2003	46.0	29.0	39.0
Burkina Faso	2003	52.4	19.2	46.4
Cameroon	2007	55.0	12.2	29.9
Malawi	2005	55.9	25.4	52.4
Tanzania	2001	38.7	29.5	35.7
Uganda	2006	34.2	13.7	31.1
Zambia	2004	72.0	53.0	68.0
Asia				
Bangladesh	2005	43.8	28.4	40.0
India	2000	30.2	24.7	28.6
Indonesia	2004	20.1	12.1	16.7
Uzbekistan	2003	29.8	22.6	27.2
Vietnam	2002	35.6	6.6	28.9
Latin America				
Bolivia	2007	63.9	23.7	37.7
Brazil	2003	41.0	17.5	21.5
Dominican Republic	2007	54.1	45.4	48.5
Guatemala	2006	72.0	28.0	51.0
Honduras	2004	70.4	29.5	50.7
Mexico	2004	56.9	41.0	47.0
Peru	2004	72.5	40.3	51.6

Source: data from World Bank, *World Development Indicators, 2010* (Washington, D.C.: World Bank, 2010), tab. 2.7.

Indigenous Poverty in Latin America

Population below the Poverty Line (%), Early 1990s			Change in Poverty (%), Various Periods		
Country	Indigenous	Nonindigenous	Period	Indigenous	Nonindigenous
Bolivia	64.3	48.1	1997–2002	0	–8
Guatemala	86.6	53.9	1989–2000	–15	–25
Mexico	80.6	17.9	1992–2002	0	–5
Peru	79.0	49.7	1994–2000	0	+3

Sources: Data for the left side of the table from George Psacharopoulos and Harry A. Patrinos, "Indigenous people and poverty in Latin America," *Finance and Development* 31 (1994): 41, used with permission; data for the right side of the table from Gillette Hall and Harry A. Patrinos, eds., *Indigenous Peoples, Poverty, and Human Development in Latin America, 1994–2004* (New York: Palgrave Macmillan, 2006).

V. Policy options on income inequality and poverty

- Areas of Intervention:
  - Altering the functional distribution
  - Mitigating the size distribution
  - Moderating (reducing) the size distribution at upper levels
  - Moderating (increasing) the size distribution at lower levels

- Policy options
  - Changing relative factor prices
  - Progressive redistribution of asset ownership
  - Progressive taxation
  - Transfer payments and public provision of goods and services

Case Study: Ghana and Cote dIvoire

- Ghana and Cote d’Ivoire
  - Land area: 92,456 and 124,502 sq miles.
  - Population in 2012: 25.5 and 20.6 millions.
  - Colonial history: 1821-1957 by Britain and 1842-1960 by France.
  - Poverty and human development in 1990: 0.393 and 0.360
  - Poverty and human development in 2013: 0.558 and 0.432
  - GDP in 1960: $594 and $ 1675
  - GDP in 2007: $1653 and $2228
  - Life expectancy in 1960: 46 and 51
  - Life expectancy in 2012: 64 and 55
  - Poverty in 1987: 46.51% and 3.28%
  - Poverty in 2006: 28.6% and 23.8%

Case Study: Ghana and Cote dIvoire

- Ghana and Cote d’Ivoire
  - Colonial impact and the legacy of institutions
  - Institutional quality
  - Ethnoliguistic fractionalization
  - Population
  - Extreme inequality
  - Common law vs civil law
  - French vs British rule
  - Education
  - Development policies